

Injective Examples

Example 1

Let $f : \mathbb{Z} \to \mathbb{Z}$ be defined by $f(x) = 3x + 2$
Assume $f(x) = f(y),\ \forall x, y \in \mathbb{Z}$
then: $3x + 2 = 3y + 2$
$3x = 3y$
$x = y$
Therefore, if $f(x) = f(y), x = y$
$=>$ $f(x)$ is injective

Example 2

Let $f : \mathbb{Z} \to \mathbb{Z}$ be define by $f(x) = x^2$
Assume $f(a) = f(b),\ \forall a, b \in \mathbb{Z}$
$then : a^2 = b^2$
$a^2 - b^2 = 0$
$(a - b)(a + b) = 0$
$a = \pm b$
a is not guranteed to be equal to b
$=>$ $f(x)$ is not injective

Deviation Method

$$\text{if } f(a) < f(b) \text{ whenever: } a < b \text{ then f is increasing}$$
$$\text{if } f(a) > f(b) \text{ whenever: } a > b \text{ then f is decreasing}$$

If we can show the first derivation is positive (i.e. f'(x) > 0) over the open interval [a, b] then we have shown f(x) is increasing.

Similarly, if we can show that the first derivation is negative (i.e. f'(x) < 0) over the open interval [a, b], then we have shown f(x) is decreasing.